DO NOW — RECALL

# Trees and binary search trees: implementation

Name: _____ Class: _____ Date: _____

**1. What does a null Next mean in a singly linked list?**

**2. In a BST, where does a smaller key go?**

**3. What is the base case of a recursive insertion reaching an empty child?**

**4. Insert 10,20,30 in that order into an empty unbalanced BST. What shape results?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

It marks the end of the list.

### 2. Model answer

Into the left subtree.

### 3. Model answer

Create and return a new node at that null position.

### 4. Model answer

A right-leaning chain 10→20→30.

### Vocabulary — know these terms

binary search tree — Binary tree in which lower keys occur to the left and higher to the right.

root — Topmost tree node.

leaf — A node with no children.

in-order traversal — Visit left subtree, node, then right subtree.

successor — Next key in in-order sorted sequence.

# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Model the reasoning

35<40 leads left, placing 35 as the left child of 40.

IMPROVE: Explain one correction to your own reasoning.





NEW LEARNING

# Read, annotate and explain

### Find the starting node

The root is 40, so compare 25 with 40 and move left.

### Repeat comparisons

25>20 goes right to 30; 25<30 goes left.

### Insert at a null child

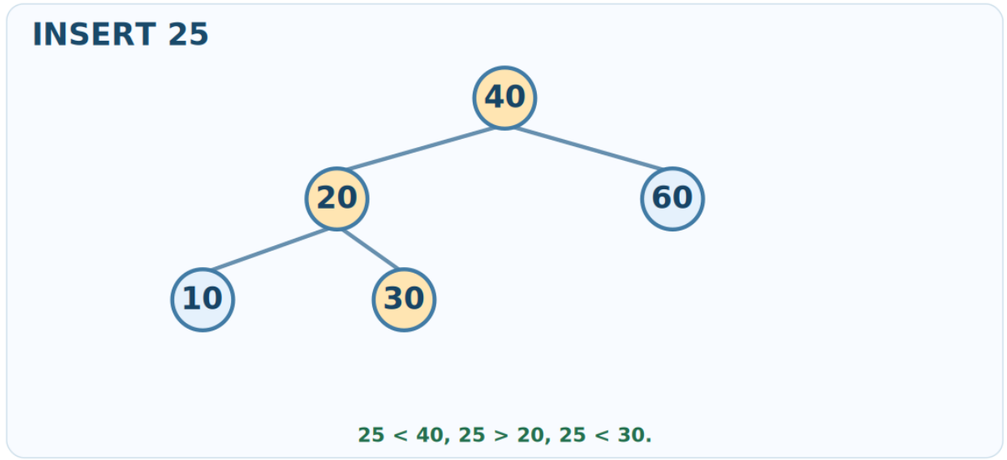
Create node 25 at the empty left-child position of 30. This implementation ignores duplicate keys: when key equals node.Key, neither subtree changes.

**Explain one step in your own words, using the correct technical terms.**

## New Learning · original lesson diagrams

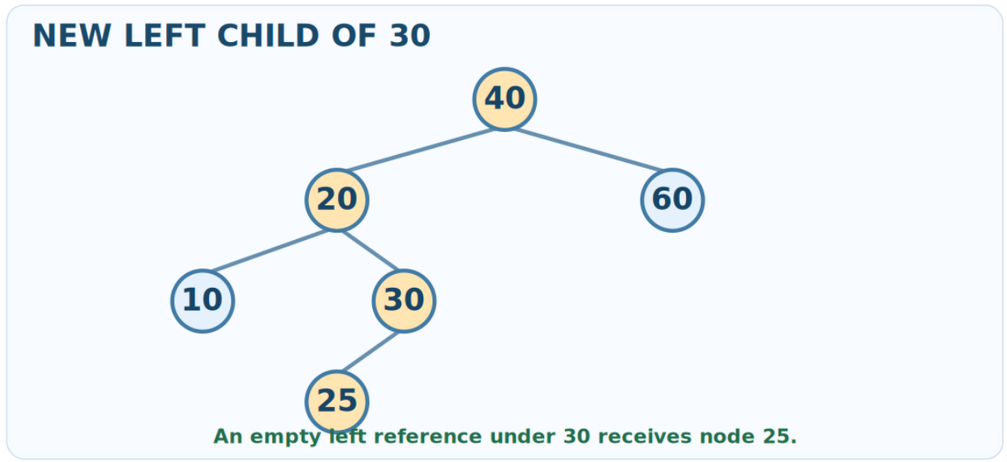
Diagrams appear after the related explanation. Annotate the original source visual.

**New Learning**




INSERT 25: 40, 20, 60, 10, 30. 25 < 40, 25 > 20, 25 < 30.

**New Learning**




NEW LEFT CHILD OF 30: 40, 20, 60, 10, 30, 25. An empty left reference under 30 receives node 25.



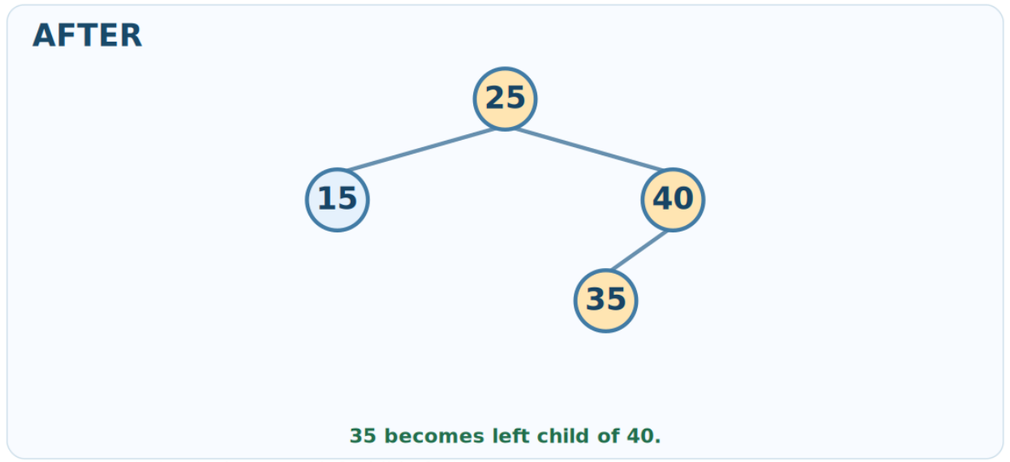


## We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Insert 35 into BST root 25, left 15 and right 40.**

35>25 leads right from the root.




AFTER: 25, 15, 40, 35. 35 becomes left child of 40.

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**In-order traversal of a BST with distinct keys visits values in ascending order.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

### GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





I DO — WORKED EXAMPLE

# Follow the worked method

**Insert 12 into the BST 20 (root), 10 (left), 30 (right).**

### Predict first

Compare 12 with 20, then 10.

### Model the reasoning

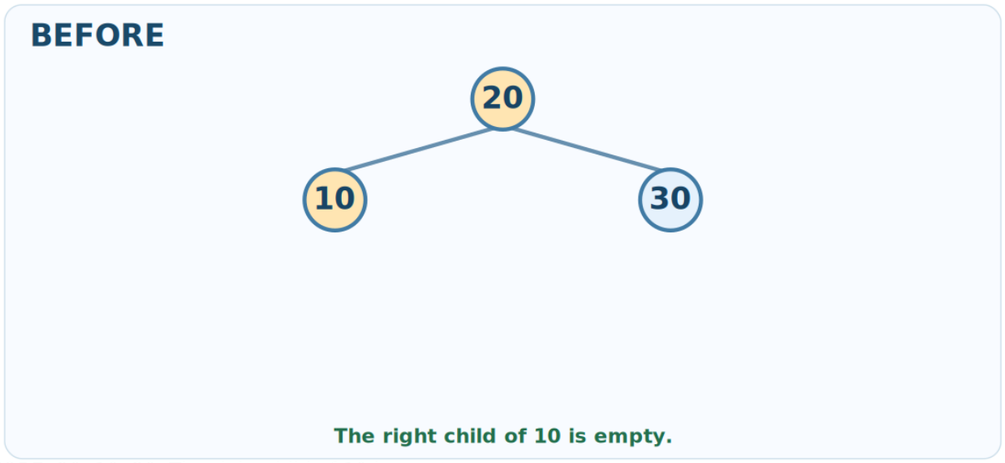
Insert 12 as the right child of 10, preserving the BST property.

# I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Insert 12 into the BST 20 (root), 10 (left), 30 (right).**

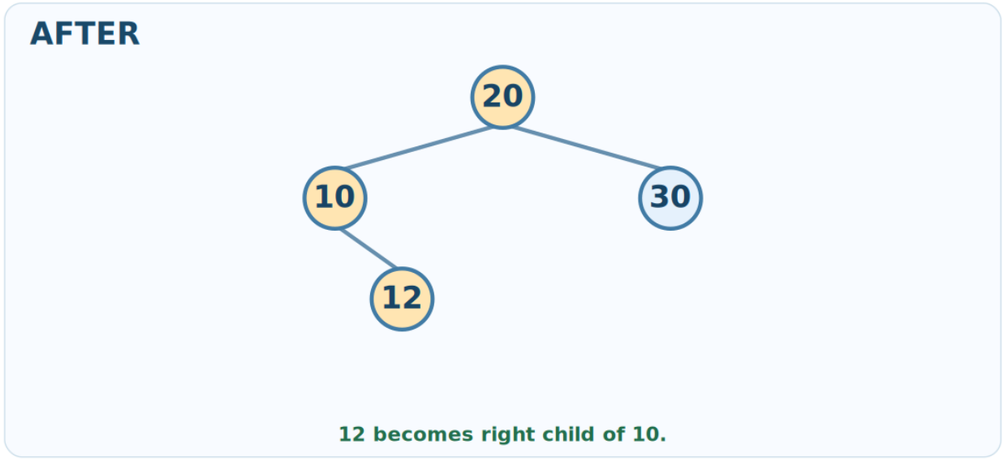
Compare 12 with 20, then 10.




BEFORE: 20, 10, 30. The right child of 10 is empty.

**Insert 12 into the BST 20 (root), 10 (left), 30 (right).**

Compare 12 with 20, then 10.




AFTER: 20, 10, 30, 12. 12 becomes right child of 10.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

In BST 40 with children 20 and 60, and 20 with children 10 and 30, delete leaf 10. Explain a second case in which deletion is more involved.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Insert 12 into BST root 15, left 9 and right 20. Identify its parent and position.

### Question 2

Insert 50,30,70,20,40,35 into an empty BST. State parent and side of 35, and the in-order visit sequence.





WE DO — GUIDED PRACTICE

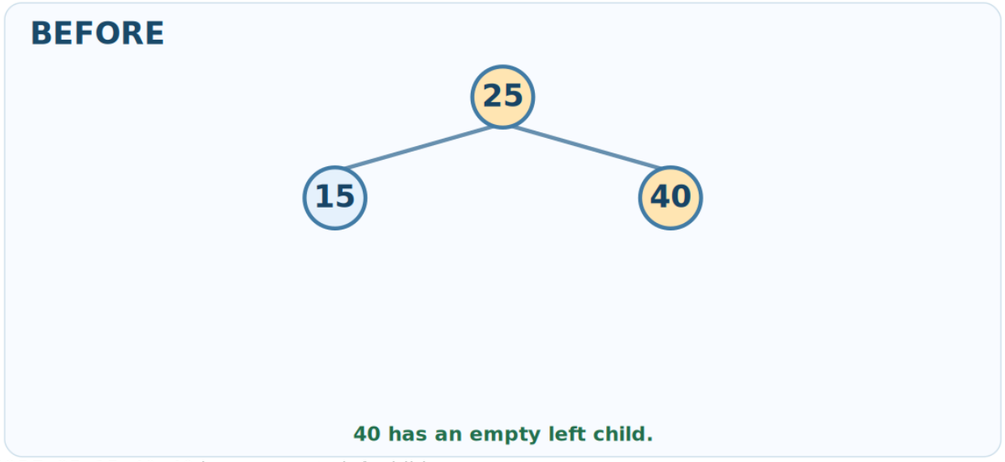
# Try → Discuss → Improve

## We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Insert 35 into BST root 25, left 15 and right 40.**

35>25 leads right from the root.




BEFORE: 25, 15, 40. 40 has an empty left child.





## WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Predict first 35>25 leads right from the root.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.